

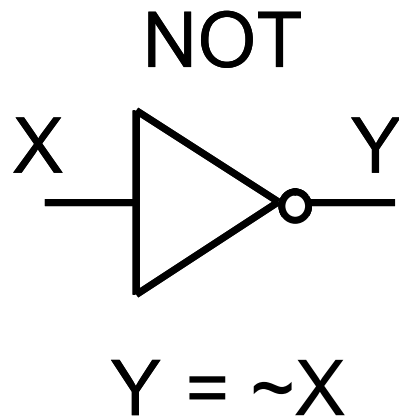
Basic Logic Gates

From Oakland University's EGR240 Course
Discussion D5.1

Basic Logic Gates and Basic Digital Design

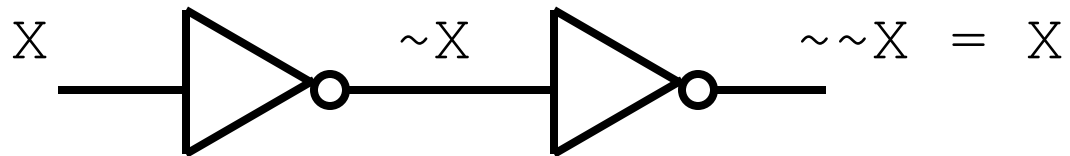
- **NOT, AND, and OR Gates**
- NAND and NOR Gates
- DeMorgan's Theorem
- Exclusive-OR (XOR) Gate
- Multiple-input Gates

NOT Gate -- Inverter



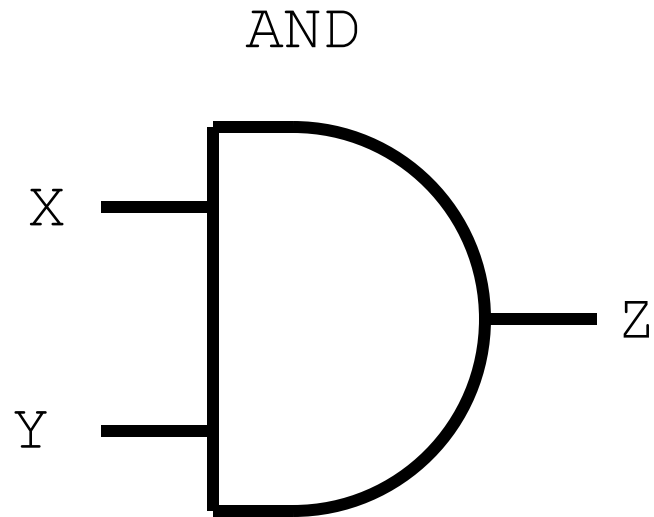
X	Y
0	1
1	0

NOT



X	$\sim X$	$\sim\sim X$
0	1	0
1	0	1

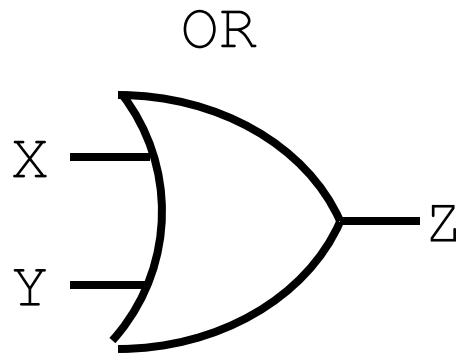
AND Gate



$$Z = X \& Y$$

X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate



$$Z = X \mid Y$$

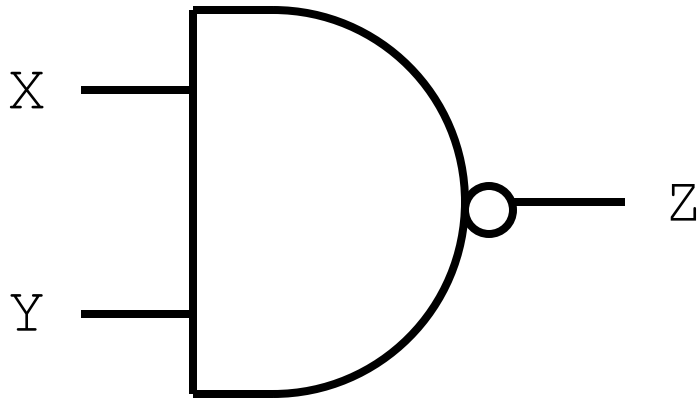
X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

Basic Logic Gates and Basic Digital Design

- NOT, AND, and OR Gates
- **NAND and NOR Gates**
- DeMorgan's Theorem
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NAND Gate

NAND

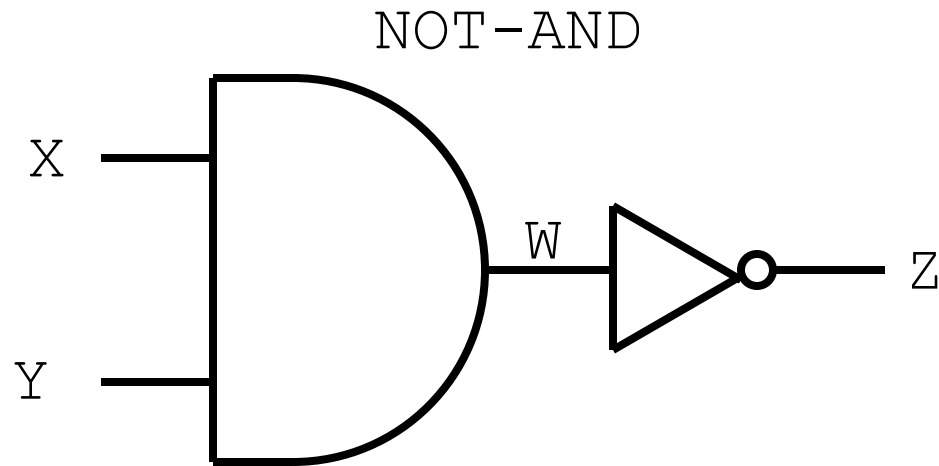


X	Y	Z
0	0	1
0	1	1
1	0	1
1	1	0

$$Z = \sim (X \ \& \ Y)$$

nand(Z, X, Y)

NAND Gate

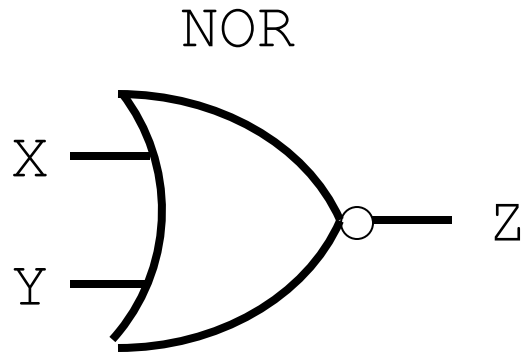


$$W = X \& Y$$

$$Z = \sim W = \sim (X \& Y)$$

X	Y	W	Z
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

NOR Gate

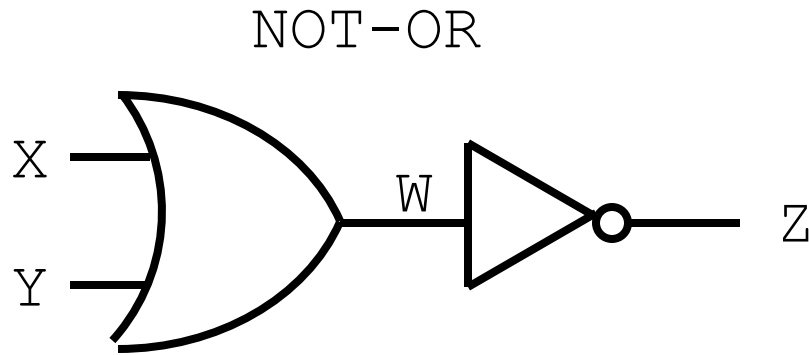


$$Z = \sim (X \mid Y)$$

nor (Z, X, Y)

X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	0

NOR Gate



$$W = X \mid Y$$

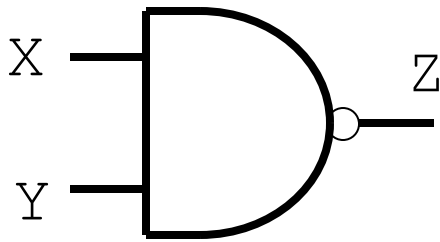
$$Z = \sim W = \sim (X \mid Y)$$

X	Y	W	Z
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

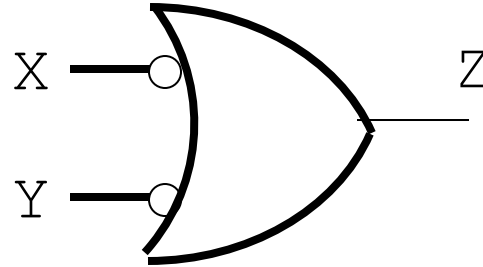
Basic Logic Gates and Basic Digital Design

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- **DeMorgan's Theorem**
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- Multiple-input Gates

NAND Gate



=



$$Z = \sim (X \ \& \ Y)$$

$$Z = \sim X \mid \sim Y$$

X	Y	W	Z
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

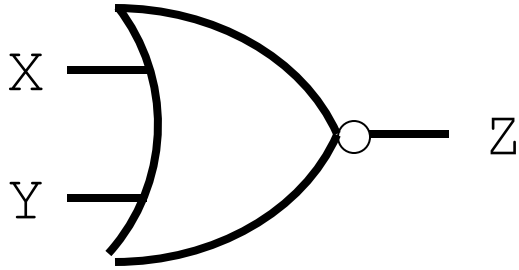
X	Y	$\sim X$	$\sim Y$	Z
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

De Morgan's Theorem-1

$$\sim (X \ \& \ Y) = \sim X \ | \ \sim Y$$

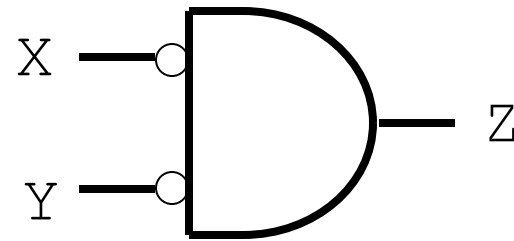
- NOT all variables
- Change & to | and | to &
- NOT the result

NOR Gate



$$Z = \sim (X \mid Y)$$

X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	0



$$Z = \sim X \ \& \ \sim Y$$

X	Y	$\sim X$	$\sim Y$	Z
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

De Morgan's Theorem-2

$$\sim (X \mid Y) = \sim X \ \& \ \sim Y$$

- NOT all variables
- Change & to $\mid$ and $\mid$ to &
- NOT the result

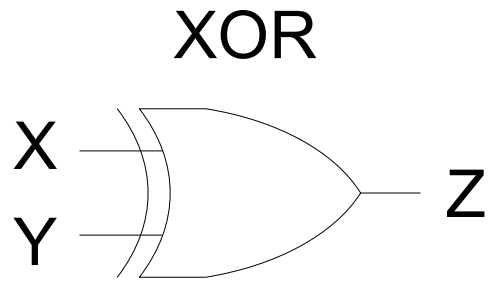
De Morgan's Theorem

- NOT all variables
- Change & to | and | to &
- NOT the result
- -----
- $\sim X \mid \sim Y = \sim(\sim\sim X \& \sim\sim Y) = \sim(X \& Y)$
- $\sim(X \& Y) = \sim\sim(\sim X \mid \sim Y) = \sim X \mid \sim Y$
- $\sim X \& !Y = \sim(\sim\sim X \mid \sim\sim Y) = \sim(X \mid Y)$
- $\sim(X \mid Y) = \sim\sim(\sim X \& \sim Y) = \sim X \& \sim Y$

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- **Exclusive-OR (XOR) Gate**
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Exclusive-OR Gate

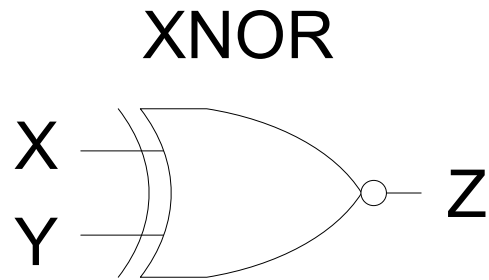


$$Z = X \oplus Y$$

xor (Z, X, Y)

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

Exclusive-NOR Gate



$$Z = \sim (X \wedge Y)$$

$$Z = X \sim \wedge Y$$

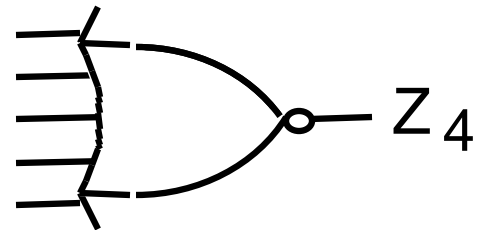
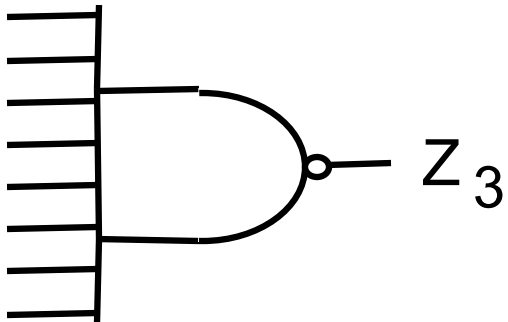
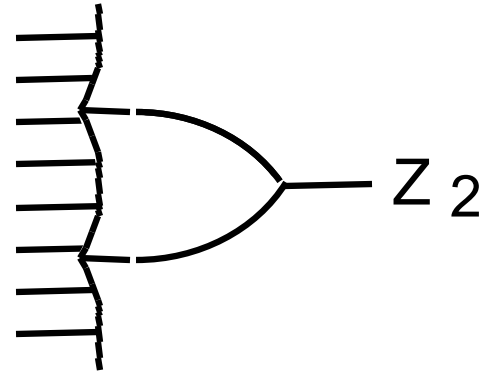
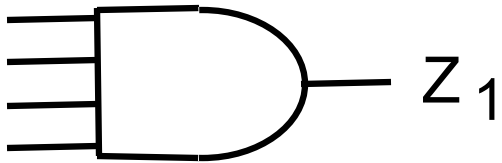
xnor (Z, X, Y)

X	Y	Z
0	0	1
0	1	0
1	0	0
1	1	1

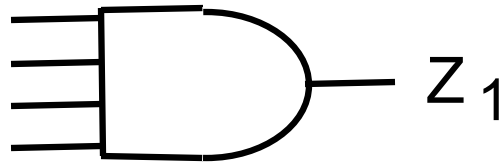
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Multiple-input Gates



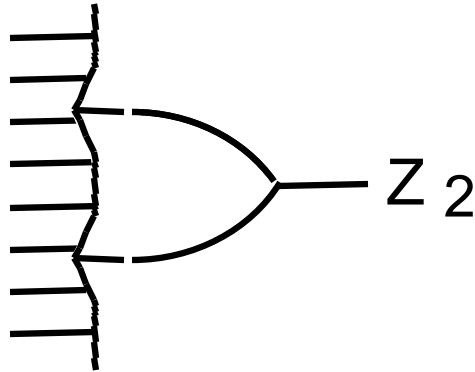
Multiple-input AND Gate



Output Z_1 is HIGH only if all inputs are HIGH

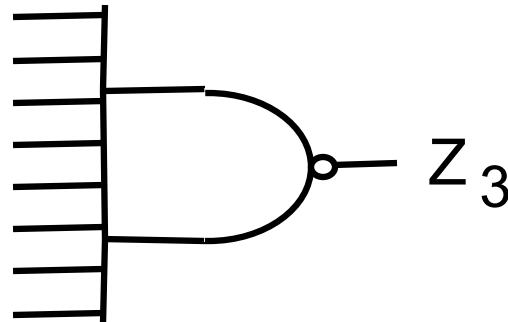
An open input will float HIGH

Multiple-input OR Gate



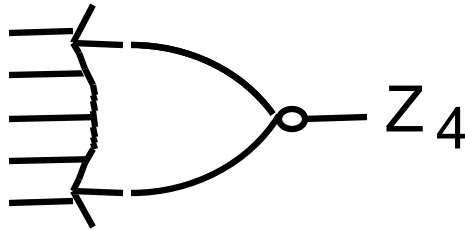
Output Z_2 is LOW only if all inputs are LOW

Multiple-input NAND Gate



Output Z_3 is LOW only if all inputs are HIGH

Multiple-input NOR Gate



Output Z_4 is HIGH only if all inputs are LOW